Decentralized Multi-Agent Bandit Learning for Intelligence Internet of Things Systems

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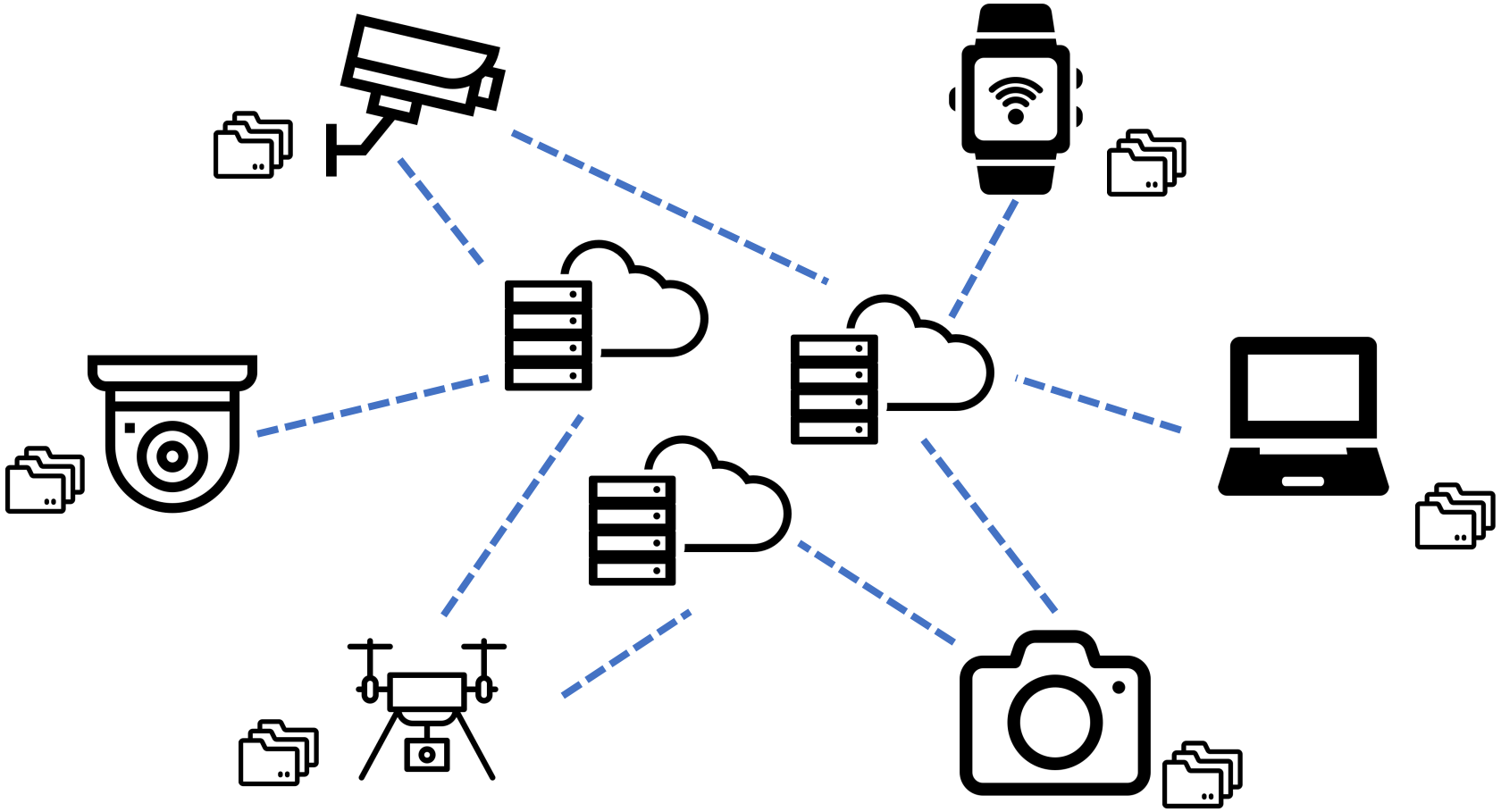
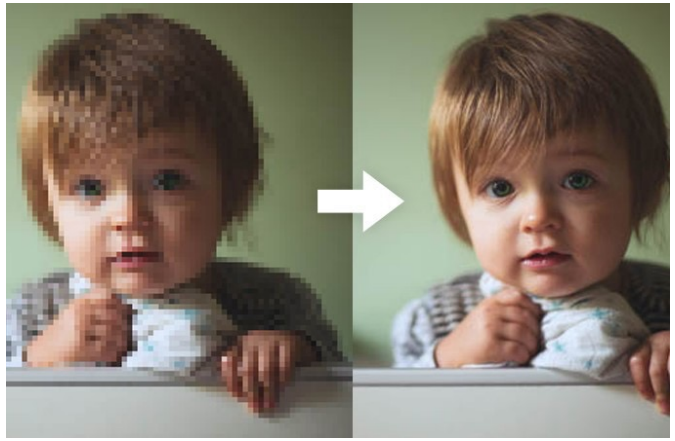
Intelligent Internet of Things Systems

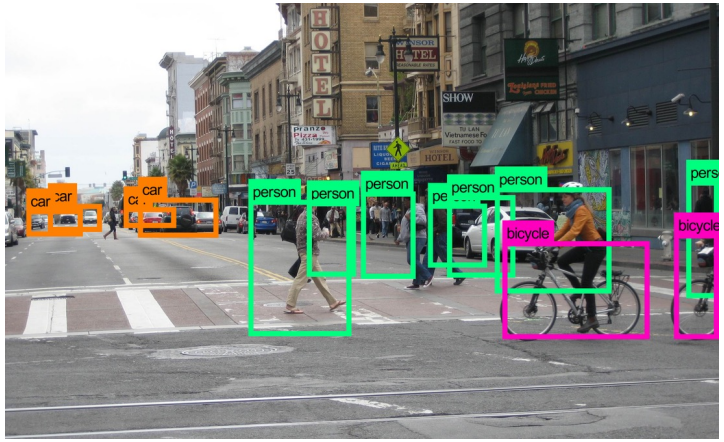


Intelligent Internet of Things Systems

Image classification



Super resolution



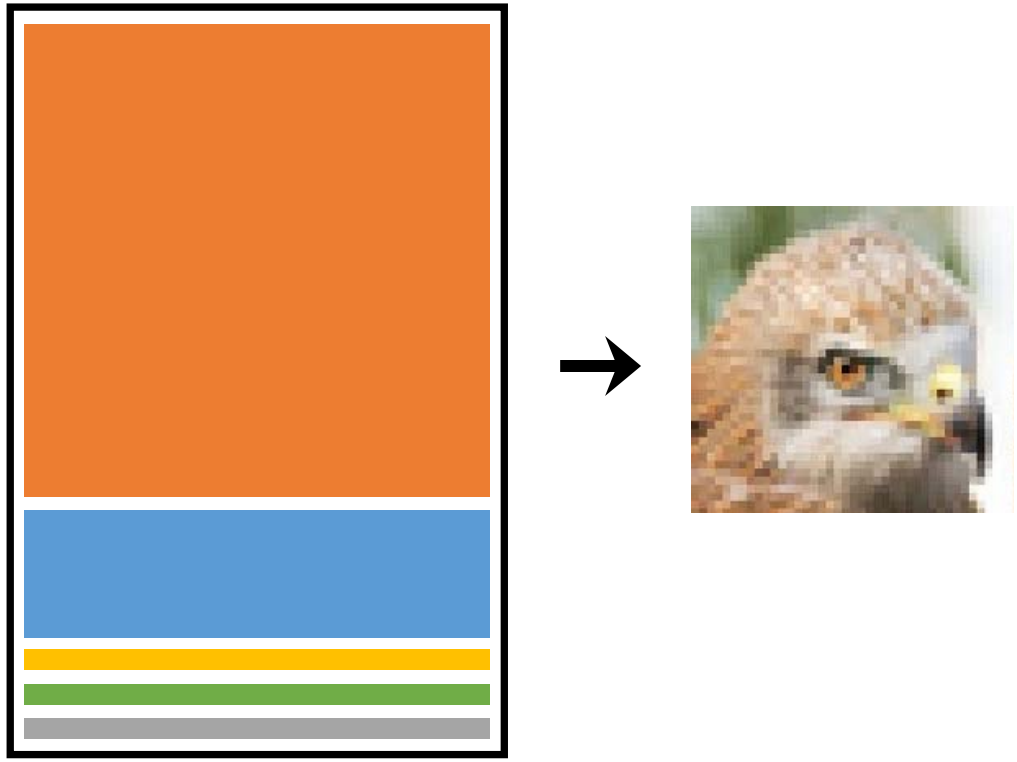
Object detection

Semantic segmentation

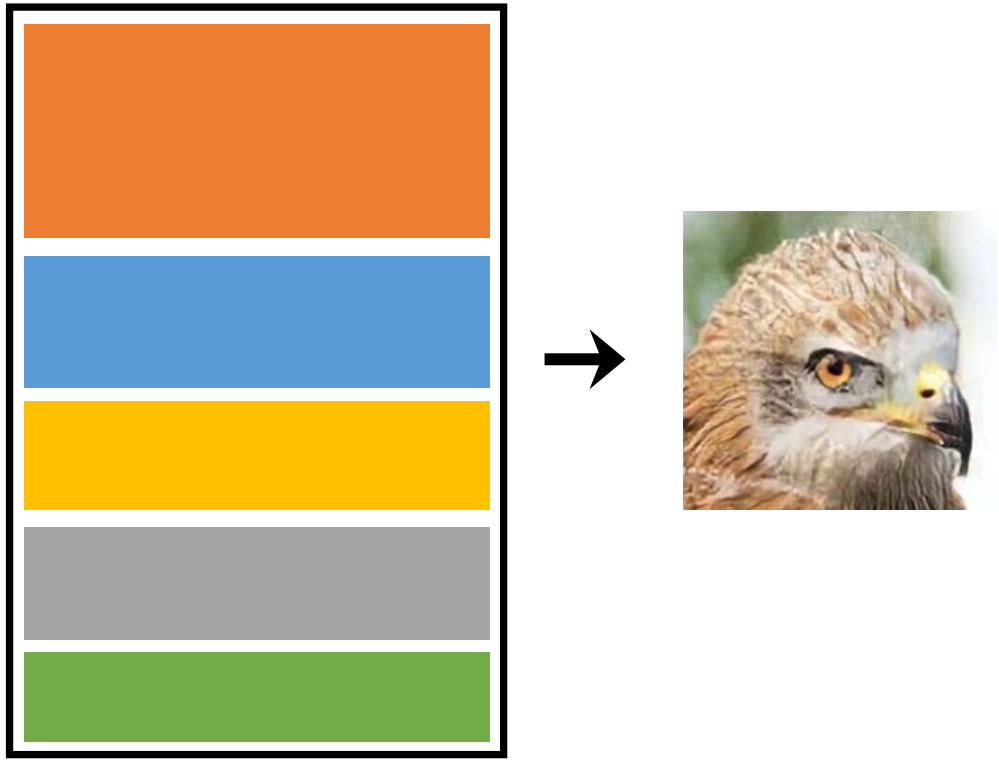
Data Dependence: Large Amount

- Amount of data $\geq 10\times$ number of parameters

Data Dependence: High Quality

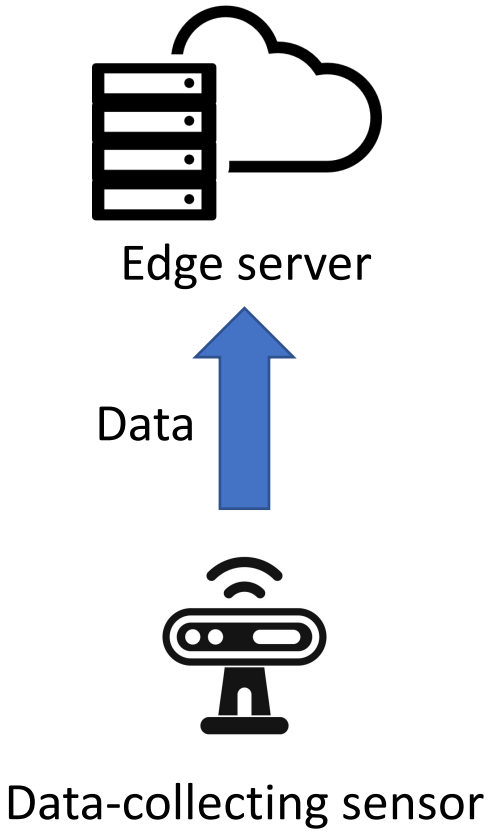


Low-quality collected data



High-quality collected data

Efficient Data Collection



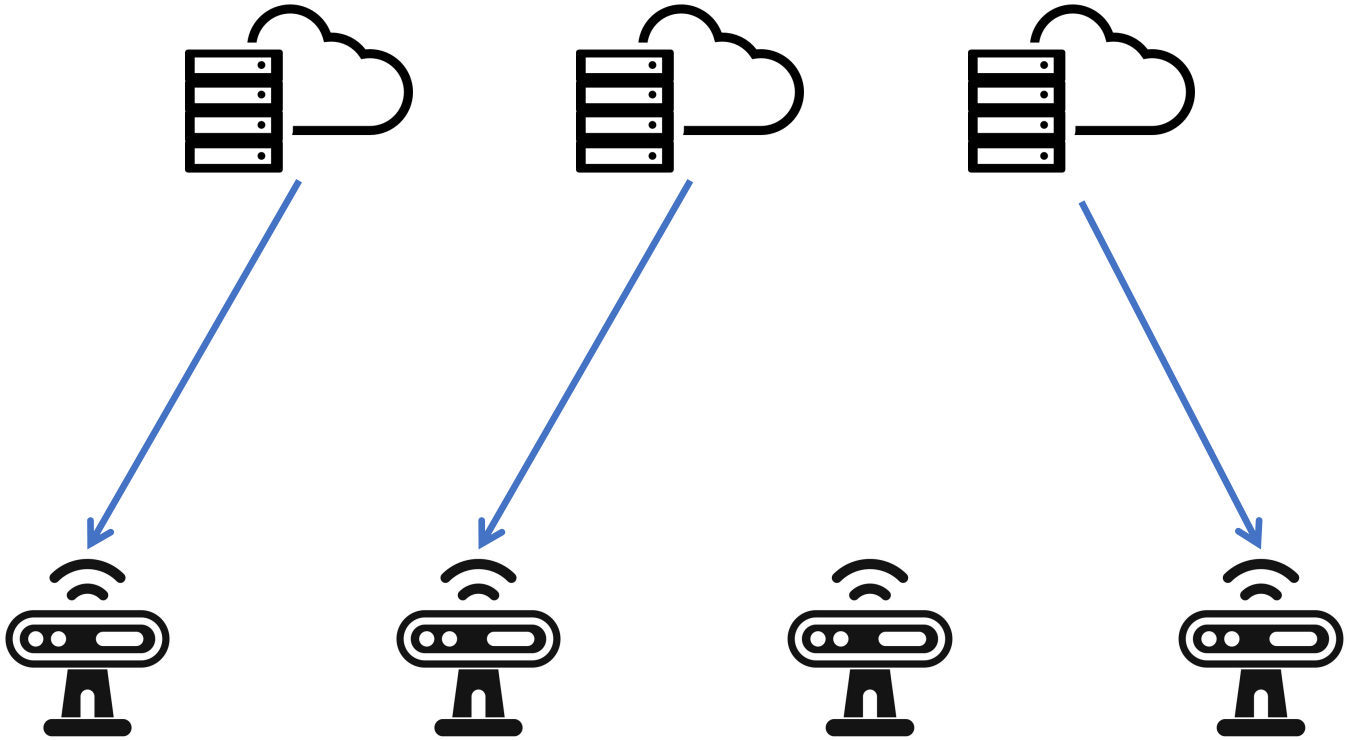
Large-amount of data

- Maximize transmission data rate

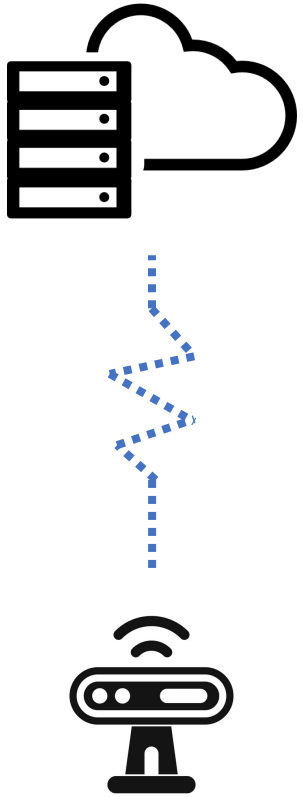
Keep data balanced in distribution

- Mitigate the impact of data heterogeneity

Decentralized Server-Sensor Association Problem

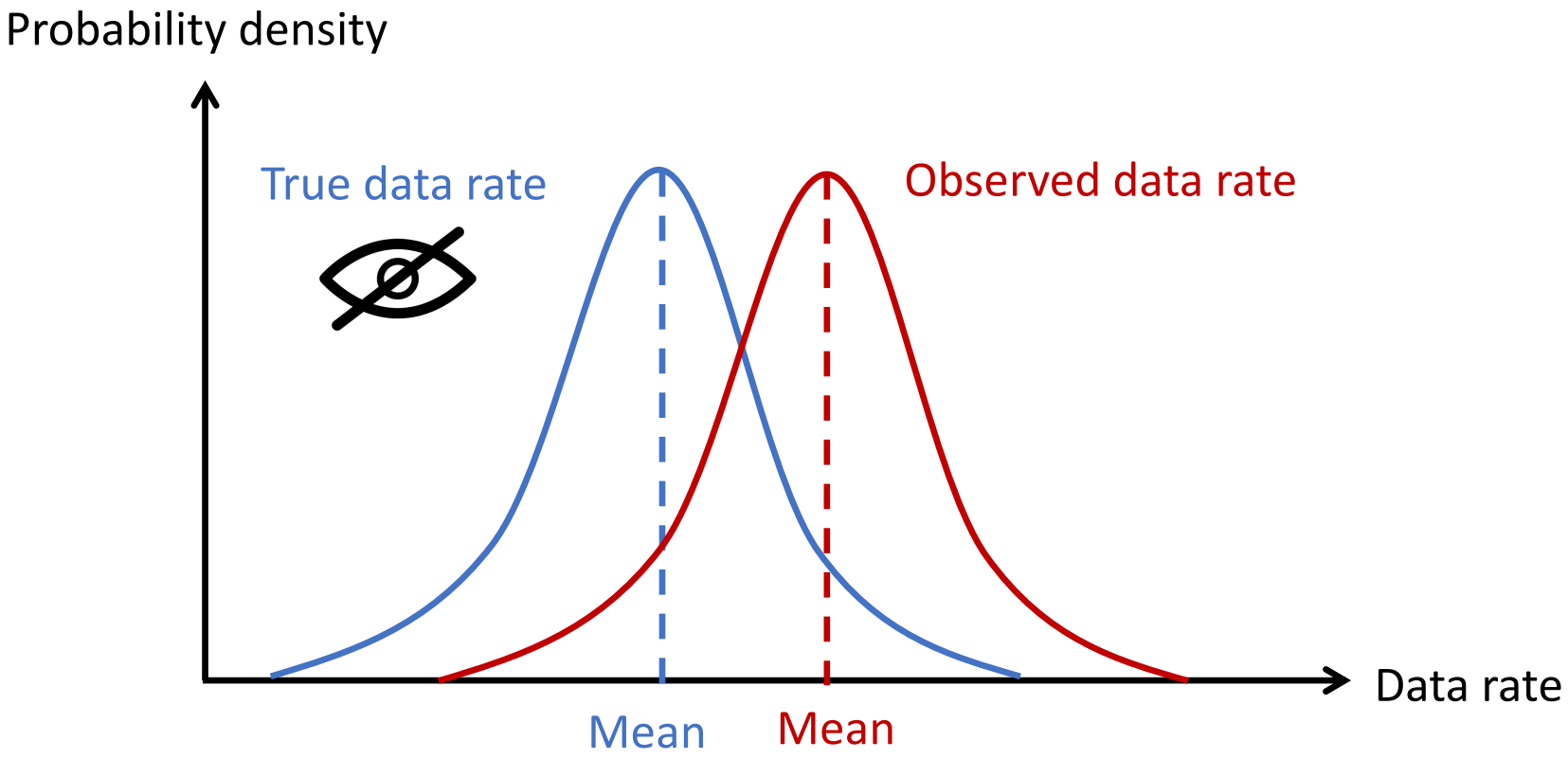


Challenge 1: Unknown Wireless Dynamics

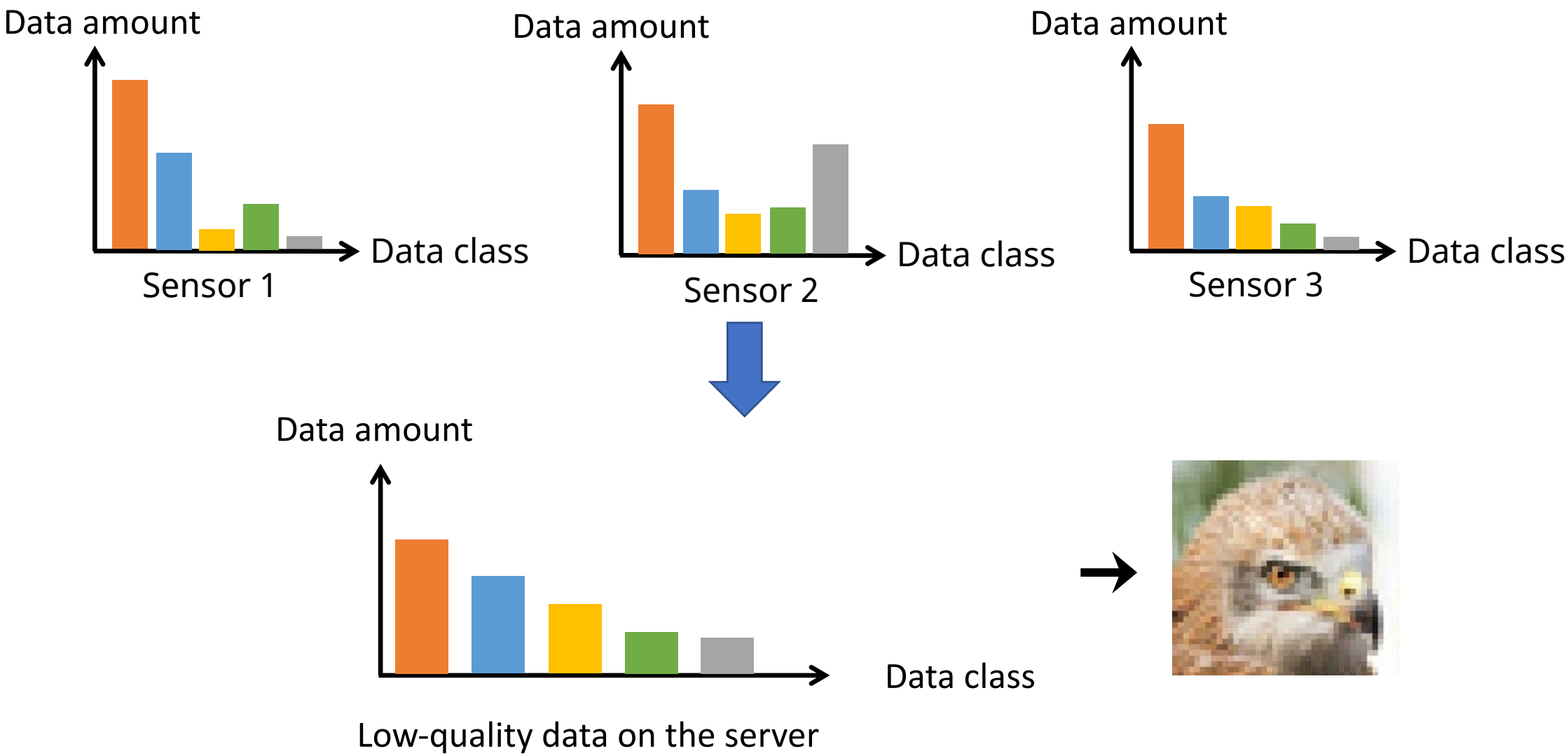


Time-varying and unknown
tranmission data rate

Challenge 2: Biased Observations for Data Rate



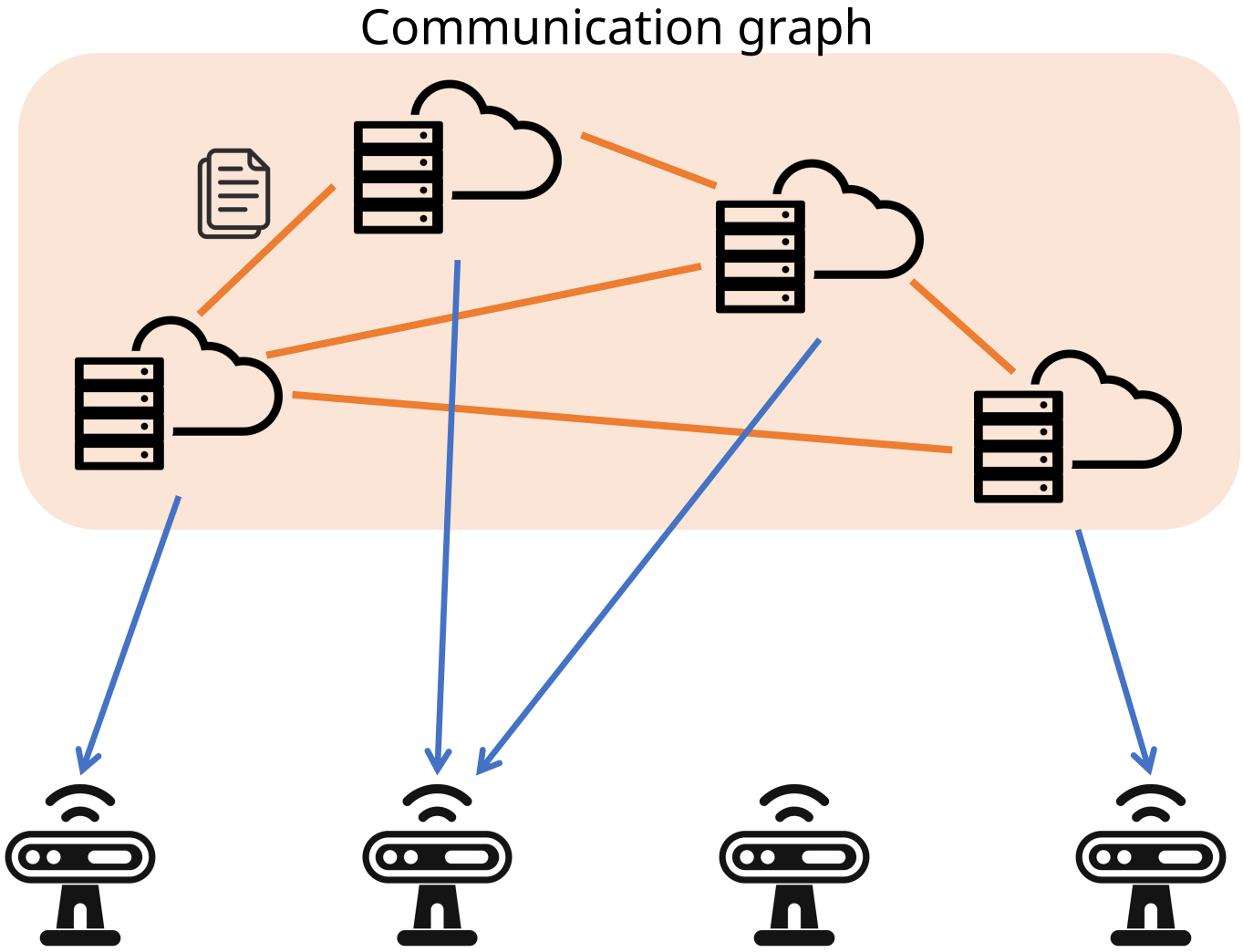
Challenge 3: Data Heterogeneity



Our Solution

- Unknown wireless dynamics → Online learning
- Biased observations for data rates → Pairwise communication
- Data heterogeneity on sensors → Online control

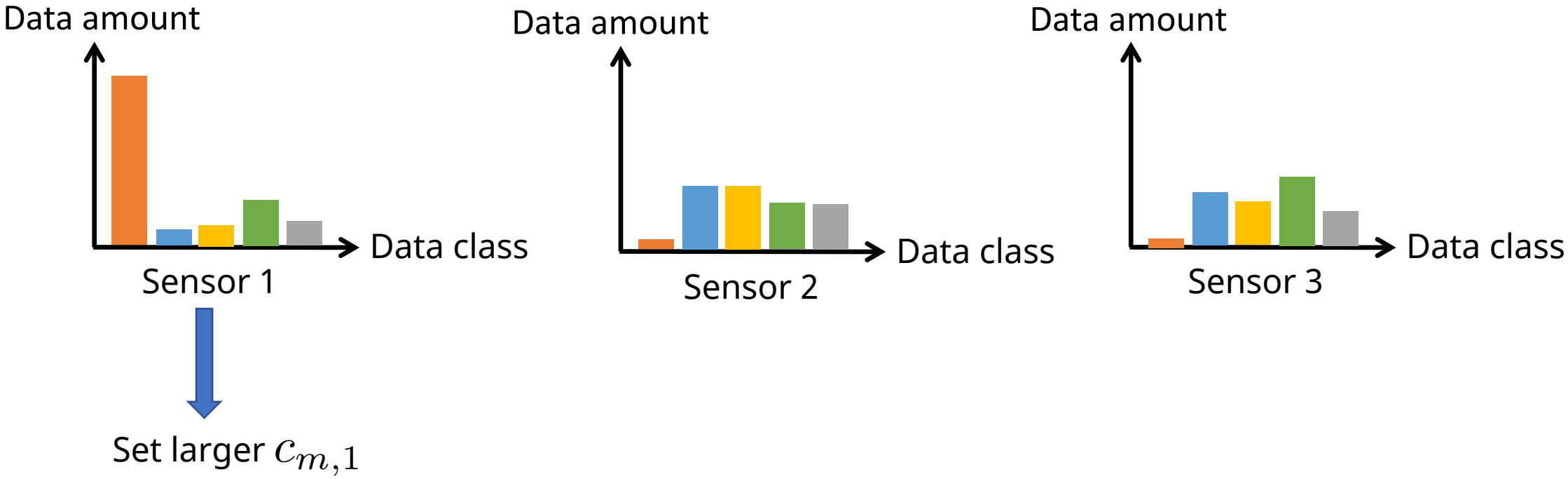
System Model: Basic Model



System Model: Data Heterogeneity

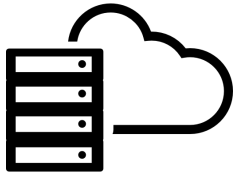
For each server m:

Time-averaged **selection fraction** constraints for each sensor $k \geq c_{m,k}$



Problem Formulation

Making server-sensor association decisions



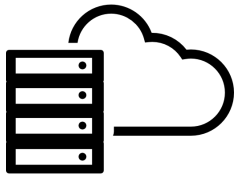
Server

Unknown before making decisions

Biased observed after making decisions

Maximize	Total true data rate
Subject to	Selection fraction constraints

Problem Reformulation : Multi-Agent Multi-Armed Bandits



Server
(Agent)

Sensor
(Arm)

Reward

Maximize Total true data rate

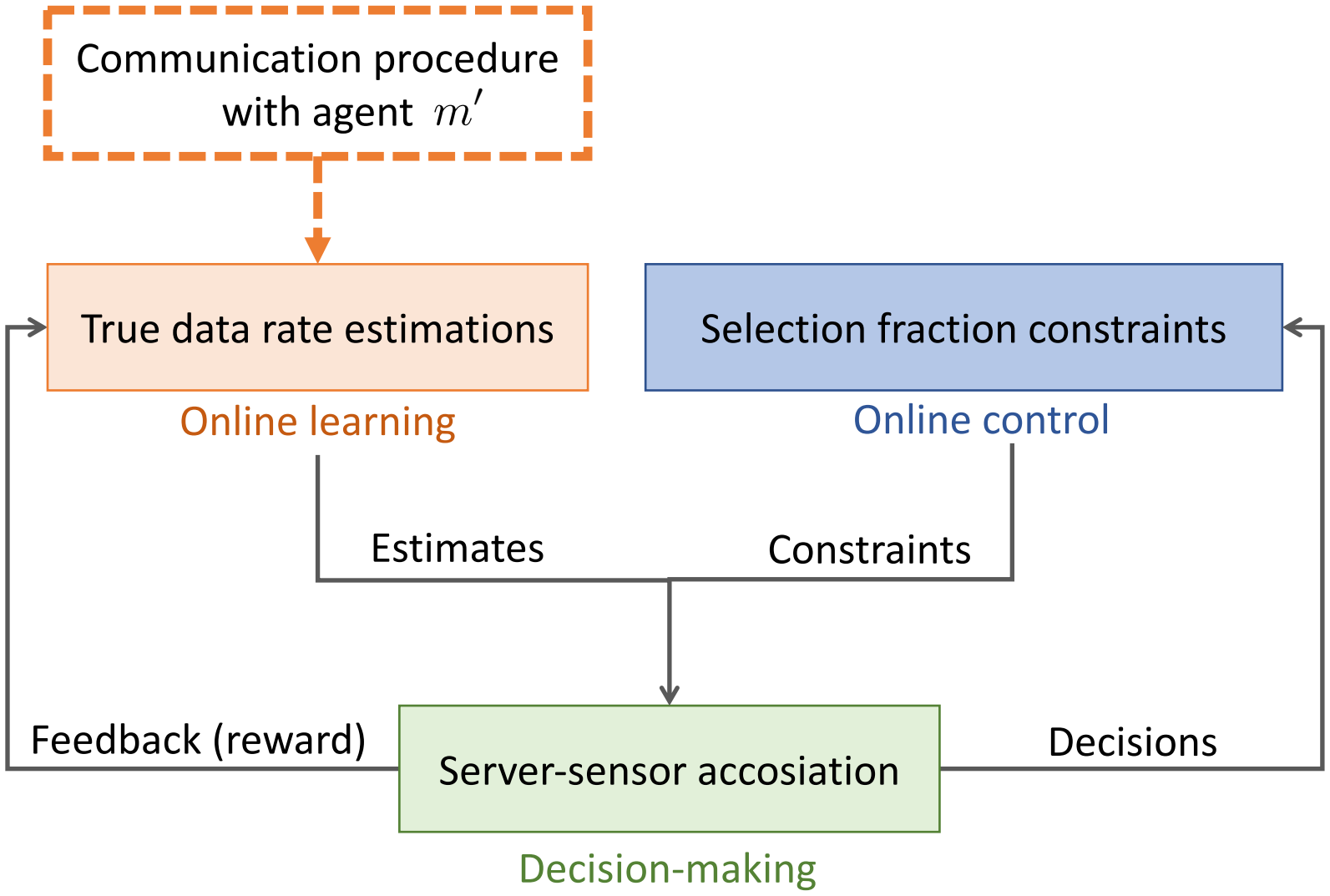
Subject to Selection fraction constraints

Regret = optimal reward - reward

Minimize Regret

Subject to Selection fraction constraints

Our Solution: DESMA



Our Solution: Online Learning under Pairwise Communication

Upper Confidence Bound (UCB) under pairwise communication estimate:

$$UCB(t) = \text{Estimate of the mean of true data rate} + C_1 \cdot \sqrt{\frac{M \log t}{\#\text{ of selected times}}} + \frac{C_2}{M^{17}}$$

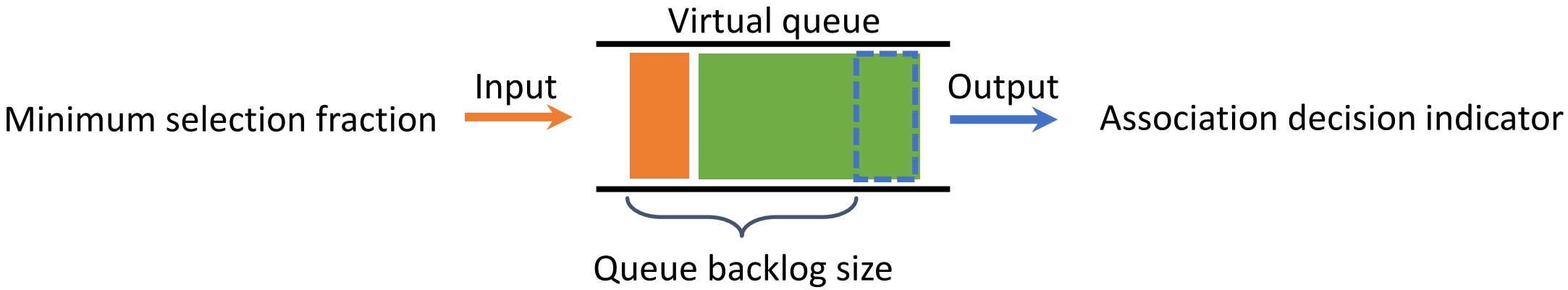
Exploitation

Exploration

Different from classical UCB due to pairwise communication

Our Solution: Online Control with Lyapunov Techniques

For each server, virtual queue for each sensor:



Selection fraction constraint is satisfied ⟷ Lyapunov optimization ⟷ Queueing process is stable

Our Solution: Integration of Online Learning and Online Control

For each server m, function $\Phi_{m,k}(t)$ for each sensor k:

$$\Phi_{m,k}(t) = \boxed{\text{Queue backlog size}} + V \cdot \boxed{\text{Estimated true data rate}}$$

For each server,

$\text{argmax}_k \ \Phi_{m,k}(t)$ ➡ Making association decision

➡ Guarantee selection fraction constraints

Performance Analysis of DESMA

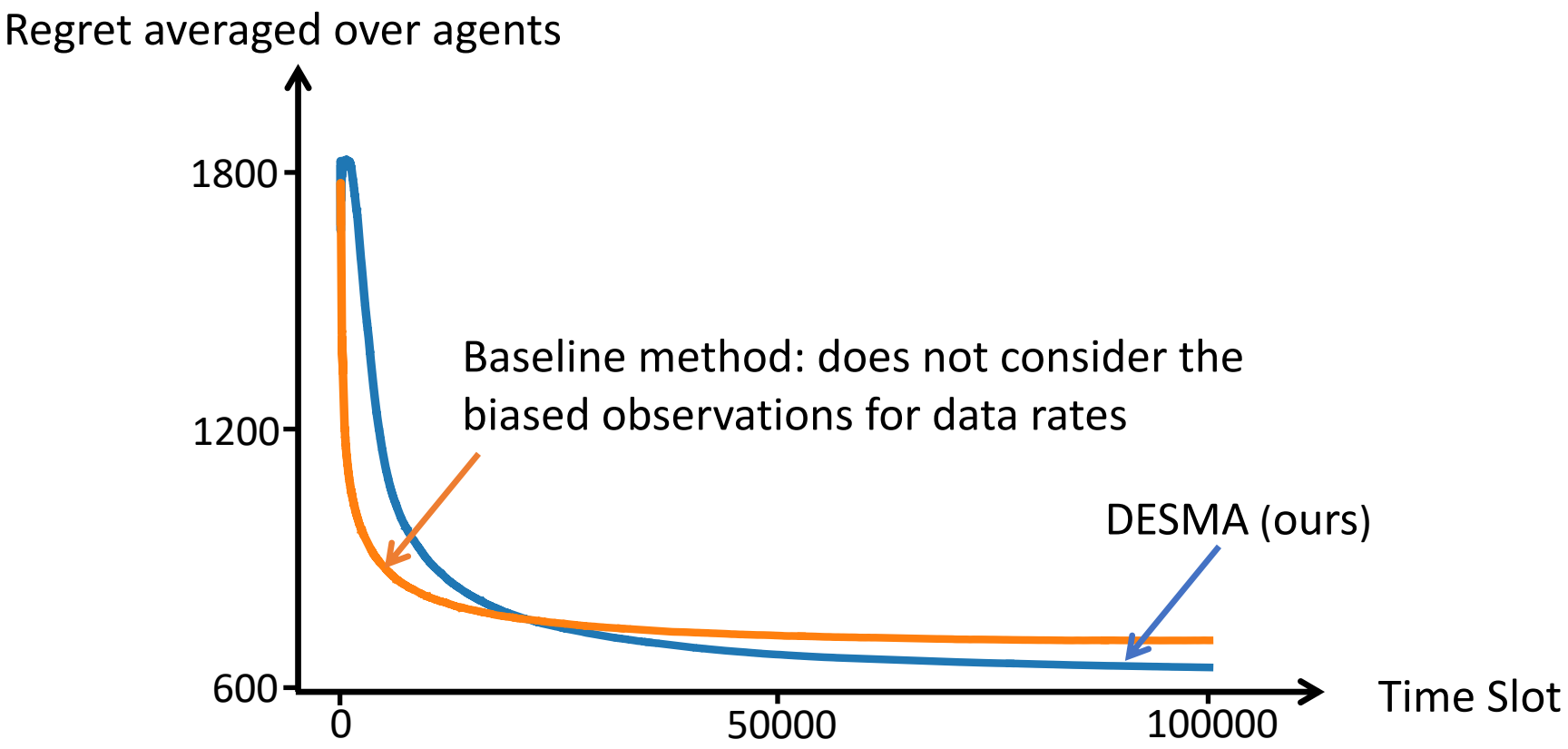
1. Time-averaged regret $\leq O(1/V) + O\left(\sqrt{(MK\log T)/T}\right)$

Control

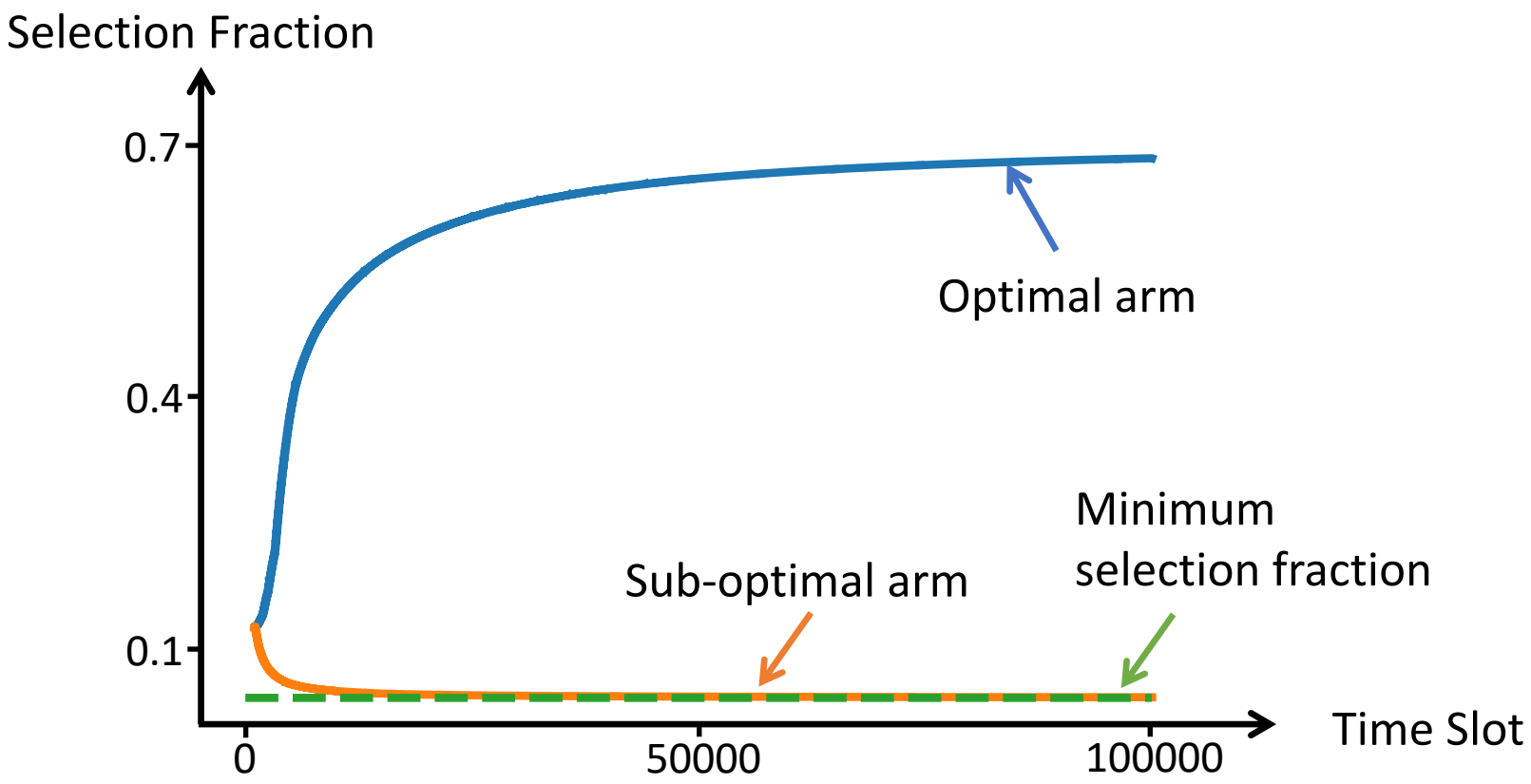
Learning

2. Selection fraction constraints are satisfied when $V < \infty$

Simulation Results: Regret

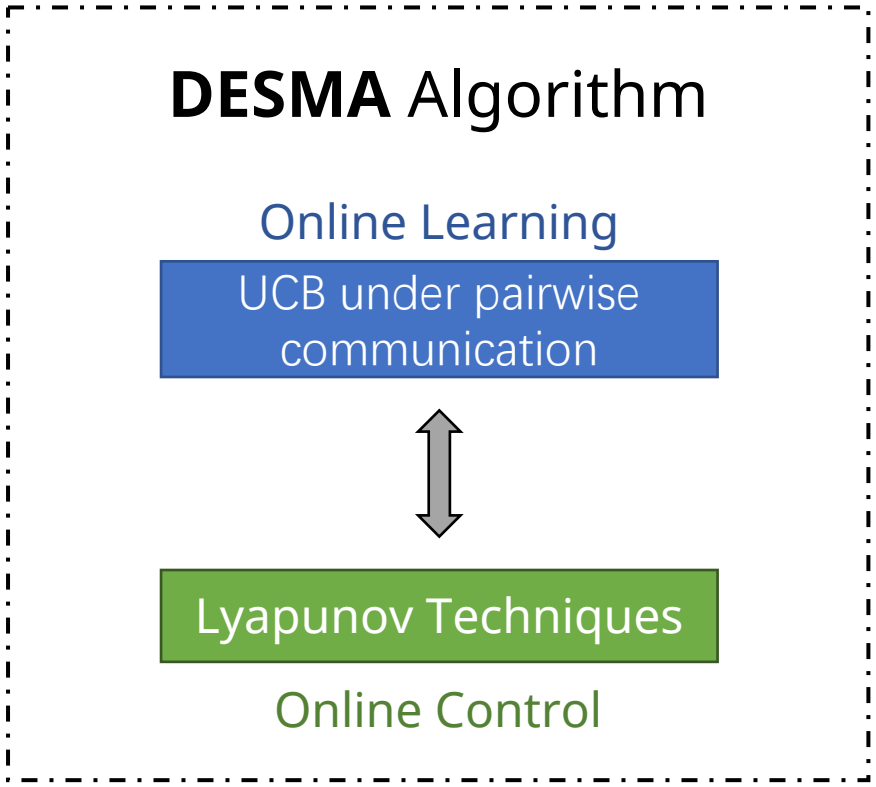


Simulation Results: Selection Fraction



Conclusion

Decentralized Server-Sensor Association with Multi-Agent Learning under Pairwise Communication



- ✓ **Sublinear** regret bound
- ✓ **Higher** data rate
- ✓ **High-quality** collected data

Thank you!

Any questions?